

Arithmetic Foundation: Powers, Exponents & Repeated Multiplication
Exam (Set 6)

Name:Class:Student ID:Time: 30 MinsTotal: 100 PtsScore:

Section 1: Write it Out! Expanded Form vs Simple Multiplication (3 pts each, 24 pts)Score: _____

Write out the expanded multiplication form and calculate the result:

Expression	Expanded Form (e.g. $a \times a \times \dots$)	True Value	Common Slip
2^3	$2 \times 2 \times 2$	8	$2 \times 3 = 6$ ✖
2^4			$2 \times 4 = 8$ ✖
3^3			$3 \times 3 = 9$ ✖
3^4			$3 \times 4 = 12$ ✖
4^2			$4 \times 2 = 8$ ✖
5^3			$5 \times 3 = 15$ ✖
10^3			$10 \times 3 = 30$ ✖

Section 2: Addition of Exponents vs Multiplication of Exponents (4 pts each, 24 pts)Score: _____

1. Simplify each expression into a single power (State why exponents are added):

(a) $10^4 \times 10^5 = 10^{+} =$ _____

(b) $2^3 \times 2^4 = 2^{+} =$ _____

(c) $3^2 \times 3^5 = 3^{+} =$ _____

(d) $10^0 =$ _____

(e) $5^1 \times 5^2 \times 5^3 = 5^{+} =$ _____

(f) $10^{-3} = \frac{1}{10^{+}} =$ _____

Section 3: Factors vs Multiples & Square Numbers Concept (4 pts each, 24 pts)Score: _____

2. [Factors of 16] A student claims: "16 can be written as eight 2s."

(a) Is the student thinking about addition or multiplication?

(b) What is the true prime factor decomposition of 16 in index notation?

(a) Identify the misconception:

(b) Prime factorization: $16 =$ _____

3. [Square Numbers Verification] Which of the following numbers are square numbers? Circle them and write their twin factors:

6: Square? Yes / No

9: Square? Yes / No

18: Square? Yes / No

25: Square? Yes / No

36: Square? Yes / No

72: Square? Yes / No

Explain why 6 is NOT a square number:

Section 4: Distributive Property vs Pure Multiplication (7 pts each, 28 pts) Score: _____

4. [Distributive Law Diagnosis] In which of the following can you use the distributive property? State YES or NO and explain:

(a) $4 \times (2.5 + 7.5)$

Distributive Property applicable? _____

(b) $4 \times (2.5 \times 7.5)$

Distributive Property applicable? _____

5. [Two-Step Word Problem] A warehouse stores boxes in stacks. Each stack has 3^2 layers, and each layer contains 2^3 boxes.

(a) Calculate how many boxes are in one stack.

(b) If there are 5 identical stacks, find the total number of boxes.



[Foundational Clinic 2] Answer Key & Scoring Guide

Section 1: Expanded Form Table (24 pts)

- $2^4 = 2 \times 2 \times 2 \times 2 = 16$

- $3^3 = 3 \times 3 \times 3 = 27$

- $3^4 = 3 \times 3 \times 3 \times 3 = 81$

- $4^2 = 4 \times 4 = 16$

- $5^3 = 5 \times 5 \times 5 = 125$

- $10^3 = 10 \times 10 \times 10 = 1000$

Section 2: Exponents Product Rule (24 pts)

1. (a) $10^{4+5} = 10^9$ (NOT 10^{20}); (b) $2^{3+4} = 2^7 = 128$; (c) $3^{2+5} = 3^7$; (d) $10^0 = 1$; (e) $5^{1+2+3} = 5^6$; (f) $\frac{1}{10^3} = 0.001$.

Section 3: Factors, Multiples & Squares (24 pts)

2. (a) Addition; (b) $16 = 2 \times 2 \times 2 \times 2 = 2^4$.

3. Squares: 9 (3×3), 25 (5×5), 36 (6×6). Not squares: 6, 18, 72. 6 is not a square because its prime factors 2×3 are different.

Section 4: Distributive Property & Applications (28 pts)

4. (a) YES: $4 \times 2.5 + 4 \times 7.5 = 10 + 30 = 40$. (b) NO: $4 \times 2.5 \times 7.5 = 10 \times 7.5 = 75$.

5. (a) $3^2 = 9$, $2^3 = 8$. One stack = $9 \times 8 = 72$ boxes. (b) 5 stacks = $72 \times 5 = 360$ boxes.